MIL-M-80177A
28 March 1974
SUPERSEDING
MIL-M-80177
9 May 1969

MILITARY SPECIFICATION

MILLING MACHINE, PLANER TYPE, DOUBLE HOUSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers double housing, planer type milling machines, with equipment and accessories as described herein.

* 1.2 Classification. The milling machines covered by this specification are of the following styles, classes and sizes. The style, class and size supplied shall be as specified (see 6.2.1), and a single siderail head shall be on the right or left side as specified.

Style 1 - One crossrail head and no siderail head

Style 2 - One crossrail head and one siderail head

Style 3 - One crossrail head and two siderail heads

Style 4 - Two crossrail heads and no siderail head

Style 5 - Two crossrail heads and one siderail head

Style 6 - Two crossrail heads and two siderail heads

Styles 1 through 6 machines include clssses and sizes as follows:

Class 1 - 20 to 30 horsepower spindle drive (light duty)

Size	Minimum Table Width, Inches
1	30
2	36
3	42
4	48

FSC 3417

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Class 2 - 30 to **100 horsepower spindle drive (medium duty)

Size	Minimum Table Width, Inches
1	42
2	48
3	54
4	60
5	66
6	72
7	78
8	84
9	90
10	96

**See Table I

Class 3 - 75 to 125 horsepower spindle drive (heavy duty)

Size	Minimum Table Width, Inches
1	44
2	56
3	62
4	68
5	72
6	80
7	92
8	104
9	116
10	140

2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

MIL-M-18058 Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

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MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards.

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies of OSHA and H28 should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS

ANS B5.1 T-Slots, Their Bolts, Nuts, Tongues and Cutters

ANS B5.18 Spindle Noses and Arbors for Milling Machines

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 West Park Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA) STANDARDS

MG1 Motors and Generators

(Copies may be obtained from the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

* 3. REQUIREMENTS

3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.2 Design. The machine shall be new and one of the manufacturer's current models. The machine shall have a bed for the purpose of supporting the table, and a horizontally mounted crossrail supported on two floor-mounted housings vertically fixed on both sides of and attached to the bed. The crossrail spindle head(s) shall be mounted vertically on the crossrail in line across the table. The siderail spindle(s) shall be mounted horizontally on the floor-mounted vertical housings. The machine axes of movement shall be (1) longitudinal: table moving along the floor-mounted bed; (2) cross: the crossrail spindle(s) moving laterally along the crossrail, and the quill of the siderail spindle(s) moving parallel with the surface of the table; (3) vertical: the crossrail and the siderail spindle head(s) moving up and down along the vertical housings and the quill of the crossrail spindle(s) moving perpendicular to the surface of the table. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

* 3.2.1 Controls. All operating controls shall be located convenient to the operator at his normal work station.

* 3.2.2 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply

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with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible. When specified (see 6.2.1) an automatic lubrication system shall be furnished.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

* 3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free of scale, mismatching, and structural defects. Processes such as welding, peening, plugging, or filling with solder or paste may be used only to improve appearance or to prevent leakage.

* 3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design except as provided in 3.3.1. None of these operations shall be employed as a repair measure for any defective part.

* 3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

* 3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

* 3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

* 3.3.6 Threads. All threaded parts shall be in the inch system and in accordance with NBS H28.

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* 3.3.7 Dials. All dials shall be graduated in the inch system. Graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch. Dial diameters shall be such that the graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

* 3.3.8 Readouts. All digital readouts shall display data in illuminated figures at least one-half inch high that give direct readings without requiring any calculation or interpolation. Each digit of the readout display shall be of the in-line type.

* 3.3.9 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.3.10 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

3.4 Components. Each milling machine furnished under this specification shall have components as described below.

* 3.4.1 Bed. The bed shall be ribbed and cross-ribbed and constructed to dimensions to insure proper alignment and support of components mounted thereon and therein. The bed shall contain the feed drive mechanism for the table and have the necessary access doors or covered openings to facilitate any required adjustments or replacement of parts. The bed shall be separately constructed, accurately machined and attached to the housings. The length of the bed and bedways shall be such that the worktable is fully supported and shall not overhang the bed in any position within its rated stroke. An adequate number of leveling screws and hold-down bolts shall be provided for the machine to be properly installed and leveled.

* 3.4.2 Housings. Housings shall be rigidly constructed with complete internal ribbing throughout their height. They shall be designed to withstand all stresses from maximum utilization of the machine within its capacity. The housings shall be fixed to the floor, and keyed or doweled and bolted to the bed. The housings shall have vertical ways for supporting the crossrail and a saddle with guideways for each siderail spindle head. The ways shall be cast integrally with the housing, or made separately with the ways aligned and securely attached to the housing. The bottom of each housing shall have an adequate number of hold-down bolts as well as leveling adjustment screws, strategically located to assure and maintain level of the machine under maximum load conditions.

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* 3.4.2.1 Housing brace. The housing brace shall have strength and rigidity to assure vibration-free performance and minimum sag, and to eliminate twisting or movement of the housings during heavy cutting loads. The housing brace shall be doweled and bolted to the housing in correct alignment. Rail elevating and rapid traverse drives shall be enclosed in the housing brace or the housing tops.

* 3.4.3 Ways and way-slides. The machine ways shall be either metallic or non-metallic type, machined, scraped and fitted. The machine shall be provided with means for adjustment to insure correct alignment of the mating components and to compensate for wear of the mating surfaces. Effective way wipers or guards shall be provided on all ways to prevent entrance of chips, dirt and other extraneous material.

* 3.4.4 Crossrail. The material and dimensions of the crossrail, and its ways matching with the milling head and housings, shall provide sufficient rigidity to withstand maximum milling forces without excessive or permanent deflection or distortion. The rail elevating mechanism shall be power operated. The crossrail shall have ways for supporting a saddle which has vertical ways for the crossrail spindle head(s).

* 3.4.5 Table. The table shall be proportioned to provide sufficient rigidity to give acceptable performance throughout the full range and capacity of the machine. The table shall be accurately machined and fitted to the bed ways. The table shall be provided with T-slots machined in accordance with ANS B5.1. The T-slots shall be parallel to the longitudinal travel of the table. Unless otherwise specified (see 6.2.1) the table shall have coolant drain channels with chip screens and provisions for returning the coolant to the reservoir. The table shall be provided with power feed and rapid traverse feeds for longitudinal table movements. An automatic control shall be provided that will stop the table drive in the event of lubrication failure.

* 3.4.5.1 Work table drive system. The table shall be driven by a gear mechanism with an adjustable anti-backlash means, driving a rack. The table drive mechanism shall be capable of smoothly accelerating, driving, decelerating and reversing the table without shock or sluggish action at the table reverse trip points, when operating at or changing from any table speed or rapid traverse speed.

* 3.4.5.2 Table controls. Unless otherwise specified (see 6.2.1), the longitudinal travel, feed rate and rapid traverse function shall be pendant controlled. The pendant control shall swing to both sides of the table to enable the operator to control the table from either side.

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* 3.4.6 Crossrail milling head(s). A crossrail milling head is defined for this specification as a self-contained assembly of the machine that houses the crossrail quill, spindle, drive motor, transmission, and associated mechanisms. The milling head(s) shall have sufficient strength and rigidity to assure accurate chatter-free performance throughout the full range and capacity of the machine. The crossrail heads shall be of the non-swiveling type. When Style 4, Style 5, or Style 6 is required, each head shall be independently controlled. The head(s) shall be provided with power and rapid traverse feed for cross movement. A power operated saddle clamp shall be provided, with a slide motor interlock device to prevent damage to the machine in the event the clamp is engaged while the head is moving. The head(s) shall be provided with an independent lubrication system. Unless otherwise specified (see 6.2.1), the head movement shall be controlled from the pendant station. The crossrail head shall have positive means for preventing rotation of the spindle when power is off.

* 3.4.6.1 Crossrail milling head(s) spindle. Each crossrail milling head spindle shall be constructed as a one-piece spindle, properly heat-treated and mounted on precision ball or roller bearings. The spindle shall have precision ground Number 50 American National Standard Taper which shall be in accordance with ANS B5.18. Unless otherwise specified (see 6.2.1) the spindle shall be provided with a power draw-bar. A graduated measuring system, without backlash, shall be provided for positioning the quill. Controls shall be furnished to start, jog and reverse the spindle. Unless otherwise specified (see 6.2.1), the quill feed and rapid traverse shall be controlled from the pendant station. When specified (see 6.2.1), the spindle speeds for Class 3 only shall be controlled from the pendant station.

* 3.4.7 Siderail milling head(s). A siderail milling head is defined for this specification as a self-contained assembly of the machine that houses the siderail spindle, quill, drive motor, transmission, and associated mechanisms. Each milling head shall have sufficient strength and rigidity to assure accurate chatter-free performance throughout the full range and capacity of the machine. The siderail milling head shall be of the non-swiveling type. The head shall be furnished with power feed and rapid traverse for vertical movement and manual adjustment to the quill. When specified (see 6.2.1), a power operated saddle clamp shall be provided, with an interlock device to prevent damage to the machine. Each head shall be provided with an independent lubrication system. When specified (see 6.2.1), the head movement shall be controlled by the operator from the pendant station. The siderail spindle head shall have positive means for preventing rotation of spindle when power is off. When specified (see 6.2.1), a ram type side head shall be furnished with not less than 16 inches of ram travel for Class 1 and 18 inches for Class 2.

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* 3.4.7.1 Siderail milling head(s) spindle. Each siderail milling head spindle shall be constructed as a one-piece spindle, properly heat-treated and mounted on precision ball or roller bearings. The spindle shall have precision ground Number 50 American National Standard Taper and be in accordance with ANS B5.18. Unless otherwise specified (see 6.2.1) the quill shall be provided with manual feed control and a manual quill clamp. A graduated measuring system, without backlash, shall be provided for positioning of the quill. Controls shall be furnished to start, stop, jog, and reverse the spindle. When specified (see 6.2.1), the quill feed, and rapid traverse shall be controlled from the pendant station. Unless otherwise specified (see 6.2.1), the spindle shall be provided with a power draw-bar. When specified (see 6.2.1), the spindle speeds for Class 3 only shall be controlled from the pendant station.

* 3.4.8 Milling head(s) and crossrail drive system. The milling head(s) and crossrail shall be driven by lead screws mounted in ball or roller bearings. The feed screws and the drive shafts shall be provided with means for backlash elimination.

* 3.4.9 Safety stops. Means shall be provided to prevent overtravel during power feed and rapid traverse of the crossrail, crossrail milling head(s) and side milling head(s). Safety devices shall also be provided to prevent damage to the feed or rapid traverse mechanisms if the moving part or component is inadvertently fed or rapid traversed into an overtravel limit switch.

3.5 Sizes and capacities. Milling machines covered by this specification shall meet the requirements of Table I. Options shall be furnished only when specified.

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TABLE I

CHARACTERISTIC	REQUIREMENT													
	Class 1				Class 2									
	Size 1	Size 2	Size 3	Size 4	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10
Table														
Working surface														
Width, inches	30	36	42	48	42	48	54	60	66	72	78	84	90	96
# Length, feet	6, 8, 10, 12, 14, 16, 18, 20, 22				10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40									
Length of stroke	Not less than table length				Not less than table length									
Feed rate range, IPM	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50
Optional range	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100	0.5 to 100
Power rapid traverse rate, IPM	200	200	200	200	180	180	180	180	180	180	180	180	180	180
Optional rate	-	-	-	-	280	280	280	280	280	280	280	280	280	280
T-slots, number of and size	5, 13/16	5, 13/16	7, 13/16	7, 13/16	5, 13/16	5, 13/16	7, 13/16	7, 13/16	7, 13/16	7, 13/16	9, 13/16	9, 13/16	9,13/16	9, 13/16
Crossrail, height from table to rail head spindle, maximum, inches	34	40	46	52	46	51	63	63	75	75	84	88	94	100
Vertical travel, inches	31	37	43	49	39	45	57	63	69	75	77	81	87	93
@ Elevation rate, range, IPM	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50
Power rapid traverse rate, IPM	30	30	30	30	30	30	30	30	30	30	30	30	30	30
Crossrail head spindles, centerline distance, inches														
At closest approach, not more than														
For 7 inch quill	16	16	16	16	-	-	-	-	-	-	-	-	-	-
For 8 inch quill	-	-	-	-	20	20	20	20	20	20	20	20	20	20
For 10 inch quill	-	-	-	-	22	22	22	22	22	22	22	22	22	22
At extreme outer travel, not less than	2 inches past opposite side of table				2 inches past opposite side of table									
Distance between siderail spindles, inches														
Quills retracted	38	44	50	56	51	57	63	69	75	81	87	93	99	105
Quills extended, not more than	24	30	36	42	31	37	43	49	55	61	67	73	79	85
Vertical travel, sideheads, inches	30	36	42	48	40	46	52	58	64	70	76	82	88	94
# Feed rate range, all heads, IPM	0.5 to 50	0.5 to 50	0.5 to 50	0.5 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50	1.25 to 50
Power rapid traverse rate, all heads, IPM	50	50	50	50	100	100	100	100	100	100	100	100	100	100
All spindles														
Spindle speeds, number of	12	12	12	12	12	12	12	12	12	12	12	12	12	12
Spindle speed range, RPM	72 to 880	72 to 880	72 to 880	72 to 880	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410	18 to 410
Optional speed ranges	36 to 880	36 to 880	36 to 880	36 to 880	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830	36 to 618 or 72 to 830
Horsepower per spindle	20	20	20	20	30	30	30	30	30	30	30	30	30	30
Optional horsepower per spindle, all styles	30	30	30	30	40, 50	40, 50	40, 50	40, 50	40, 50	40, 50	40, 50	40, 50	40, 50	40, 50
Optional horsepower per spindle, Styles 1 and 2 only	-	-	-	-	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100	60,75,100
Quill diameter, inches	7	7	7	7	8	8	8	8	8	8	8	8	8	8
Optional quill diameter	-	-	-	-	10	10	10	10	10	10	10	10	10	10
Quill travel, inches	7	7	7	7	10	10	10	10	10	10	10	10	10	10
Optional travel	-	-	-	-	14	14	14	14	14	14	14	14	14	14
Quill feed rate range, IPM	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31	0.50 to 31
Quill feed rapid traverse, IPM	40	40	40	40	40	40	40	40	40	40	40	40	40	40
Distance between housings, inches	44	50	56	62	62	66	72	78	84	90	96	102	108	144

@ Infinitely variable

As specified (see 6.2.1)

Requirements are not less than those shown, except as otherwise indicated. Where a range is shown, requirements are from the smaller figure or less to the larger figure or more.

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TABLE I

CHARACTERISTIC	REQUIREMENT									
	Class 3									
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10
Table										
Working surface										
Width, inches	44	56	62	68	72	80	92	104	116	140
# Length, feet	10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40									
Length of stroke	NOT LESS THAN TABLE LENGTH									
Feed rate range, IPM	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125	0.5 to 125
Power rapid traverse rate, IPM	180	180	180	180	180	180	180	180	180	180
Optional rate	300	300	300	300	300	300	300	300	300	300
T-slots, number of and size	5, 13/16	7, 13/16	7, 13/16	7, 13/16	9, 13/16	9, 13/16	9, 13/16	9, 13/16	9, 13/16	9, 13/16
Crossrail Height, table to rail head spindle, inches max	52	64	75	76	82	88	100	112	124	124
Vertical travel, inches	45	51	63	69	75	81	93	105	117	117
@ Elevation rate, range, IPM	0.125 to 50	0.125 to 50	0.125 to 50	0.125 to 50	0.125 to 50	0.1 to 40	0.1 to 40	0.1 to 40	0.1 to 40	0.1 to 40
Power rapid traverse rate, IPM	50	50	50	50	50	50	50	50	50	50
Crossrail head spindles, centerline distance, inches										
At closest approach, not more than										
For 10 inch quill	22	22	22	22	22	22	22	22	22	22
For 12 inch quill	25	25	25	25	25	25	25	25	25	25
For 14 inch quill	25	25	25	25	25	25	25	25	25	25
At extreme outer travel, not less than	2 INCHES PAST OPPOSITE SIDE OF TABLE									
Distance between siderail spindles, inches										
Quills retracted	53	65	71	77	83	91	103	115	127	150
Quills extended, not more than	25	37	43	49	53	63	75	87	99	124
Vertical travel, sideheads, inches	40	52	58	64	70	76	88	100	112	135
@ Feed rate range, all heads, IPM	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125	0.50 to 125
Power rapid traverse rate, all heads, IPM	100	100	100	100	100	100	100	100	100	100
All spindles										
Spindle speeds, number of	12	12	12	12	12	12	12	12	12	12
Spindle speed range, RPM	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500	10 to 500
Optional speed ranges	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833	14 to 700 or 28 to 833
Horsepower per spindle	75	75	75	75	75	75	75	75	75	75
Optional horsepower per spindle	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125	100 or 125
Quill diameter, inches	10	10	10	10	10	10	10	10	10	10
Optional quill diameter	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14	12 or 14
Quill travel, inches	14	14	14	14	14	14	14	14	14	14
Quill feed rate range, IPM	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31	0.078 to 31
Quill feed rapid traverse, IPM	40	40	40	40	40	40	40	40	40	40
Distance between housings, inches	61	73	79	85	91	99	111	123	135	164

@ Infinitely variable
As specified (see 6.2.1)
Requirements are not less than those shown except as otherwise indicated. Where a range is shown, requirements are from the smaller figure or less to the larger figure or more

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* 3.6 Performance. Each machine shall be capable of satisfactory performance when tested in accordance with 4.5.3 through 4.5.4.1.

* 3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of Table II.

TABLE II

ALIGNMENT TOLERANCES	
TESTS	TOLERANCES
Table:	
Longitudinal rise and fall	± 0.0005 in any 24 inch span
Total	0.00015 more for each additional 24 inches
Crossrail ways parallel with table travel, one or first head	± 0.0008 in any 24 inch span
Total	0.0001 more for each additional 24 inches
Second head, if applicable	± 0.0012 in any 24 inch span
Total	0.00015 more for each additional 24 inches
T-slots parallel to table travel	±0.0005 in any 24 inch span
Total	0.0001 more for each additional 24 inches
Crossrail Milling Head:	
Spindle end periphery runout	0.0005 TIR
Spindle face runout	0.0003 TIR
Spindle end taper runout	
1 inch from spindle nose	0.0005 TIR
12 inches from spindle nose	0.001 TIR
Spindle square with table	± 0.0005 in 12 inches
Spindle travel square with table	± 0.0005 in 12 inches
Siderail Milling Head:	
Spindle end periphery runout	0.0005 TIR
Spindle face runout	0.0003 TIR
Spindle end taper runout	
1 inch from spindle nose	0.0005 TIR
12 inches from spindle nose	0.001 TIR
Spindle square with motion of table	± 0.0005 in 12 inches

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3.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to JIC Standards for General Purpose Machine Tools except that a single disconnecting means shall disconnect all ungrounded conductors. The machine shall operate on 230/460 volt, 3 phase, 60 Hz power, and unless otherwise specified (see 6.2.1), shall be initially wired for operation on 230 volts (see 6.2.1). An identified terminal shall be provided suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards.

3.8.1 Motors. The spindle drive motors shall be single or variable speed. Unless otherwise specified (see 6.2.1), their duty rating shall be not less than 60 minutes. Motors shall be either AC, 3 phase, 60 Hz, or industrial DC conforming to NEMA Standard MG-1. Rectifiers for DC motors shall be solid state. Unless otherwise specified (see 6.2.1) each motor shall meet the requirements of a dripproof enclosure.

3.9 Coolant system. A coolant system shall be furnished. It shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit the full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.10 Hydraulic system. A hydraulic system, if furnished, shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Over-pressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.11 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. Electrical load indicator. The machine shall be equipped with a meter to indicate the electrical load on each spindle drive motor. The meter shall be current actuated. The dial shall indicate either amperes or percent of rated full load and shall be clearly marked to indicate the maximum rated load for operation of the machine. The instrument shall be clearly readable from the operator's normal work position.

3.12 Optional equipment. When required, the following equipment and accessories shall be furnished as specified (see 6.2.1).

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3.12.1 Chip removal system. When specified, a chip removal system shall be furnished to transport chips from the cutter area to a specified collection area.

3.12.2 Spindle elapsed time meter. When specified, the machine shall be equipped with a spindle elapsed time meter to record the accumulative running time during operation of the system, up to at least 9999.9 hours.

3.12.3 Mist coolant. When specified, the machine shall be furnished with a mist coolant system, either in addition to or in lieu of the system required in 3.9. The system shall be adequate for all types of machining and shall have means for controlling the flow of coolant. Troughs, drain plugs, pumps, piping, control nozzles and sump shall be furnished, as well as all features necessary for draining and cleaning.

3.12.4 Individual position readout. When specified, the machine system shall have individual axis position readouts. The position readouts shall be capable of readings displaying full travel in integers and decimal values for each axis.

3.12.5 Quill readout. When specified, the quill shall be furnished with an electronic position readout with precision rack and pinion pickup to monitor the movement of the quill.

3.13 Additional accessories and equipment. Additional accessories, equipment or attachments shall be furnished as specified (see 6.2.1).

3.14 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class II B equipment.

3.15 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.16 Lubrication chart or plate. Unless otherwise specified (see 6.2.1) a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Military or Federal Specification (if available)

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3.17 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Size (inches)

Power input (volts, total amps, phase, frequency)

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

3.18 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.19 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified in the contract or order. The supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the inspection when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.4.1, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.19.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of 3.7.

4.5.3 Test material for cutting tests. Unless otherwise specified (see 6.2.1), the cutting test shall be performed on four blocks of SAE 1045 steel. The blocks may be drilled to facilitate clamping. The size of the blocks shall be not less than 14 inches long, 6 inches wide and 6 inches thick. Scrap blocks may be utilized if available. Each test block shall be ground flat on one vertical side and one horizontal side.

4.5.4 Metal removal and tool chatter test. A metal removal and tool chatter test by means of roughing cuts on the test blocks shall be made at sufficient depth and feed rate to load each spindle drive motor to its maximum rated electrical input. The horizontal side of each test block that has been previously ground shall be mounted on the table surface with the vertical ground side opposite the siderail spindle cutter and in precise alignment with the nearest exposed T-slot. One block shall be mounted in each quadrant of the table. In test of multi-spindle machines, the cuts of all vertical spindles shall be made simultaneously. Horsepower of any spindle may vary from the rated value by plus or minus 5% provided the total horsepower of all spindles is not less than total rated horsepower. If structure of the machine permits simultaneous cuts with side spindles, these may also be made at the same time as the horizontal

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cuts at the option of the manufacturer. Otherwise, a maximum horsepower cut with the side spindles shall be made separately from the cuts made with the crossrail spindles. The cutter path of each spindle shall be made for the various styles of machines as follows:

Style 1 - Figure 2

Style 2 - Figure 2 and Figure 3

Style 3 - Figure 2 and Figure 3

Style 4 - Figure 1

Style 5 - Figure 1 and Figure 3

Style 6 - Figure 1 and Figure 3

The electrical load of the machine during the test shall be measured using any instrument(s) with accuracy within 2%. The machine shall be operated at a spindle speed not more than 10% of the high value of the range required under Table I for the machine funished. Except as limited in this paragraph, type of cutting tool, table feed rate, and spindle speed shall be determined by the supplier. The resultant metal removal rate shall be not less than 0.65 cubic inch per minute per rated horsepower of each spindle drive motor. The surface of the workpiece shall show no evidence of tool chatter.

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Fig. 1

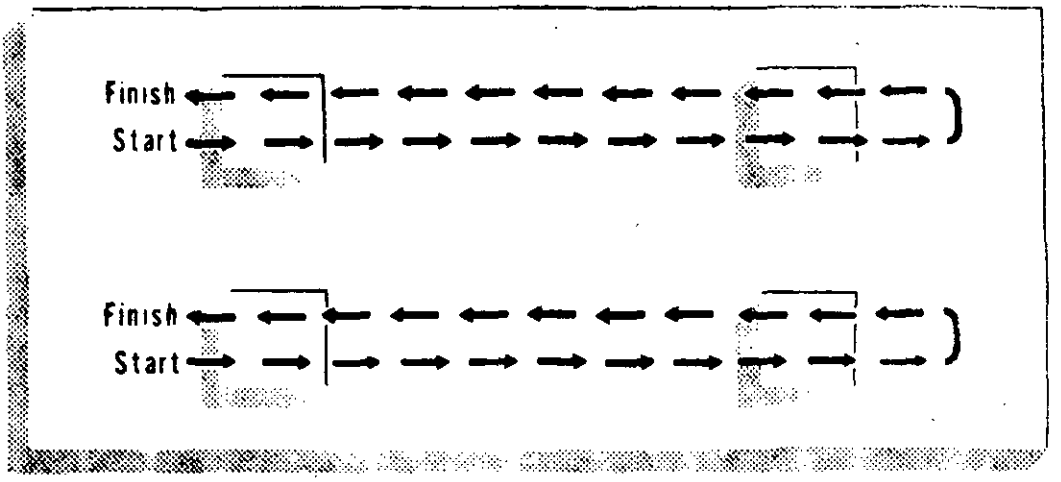


Fig. 2

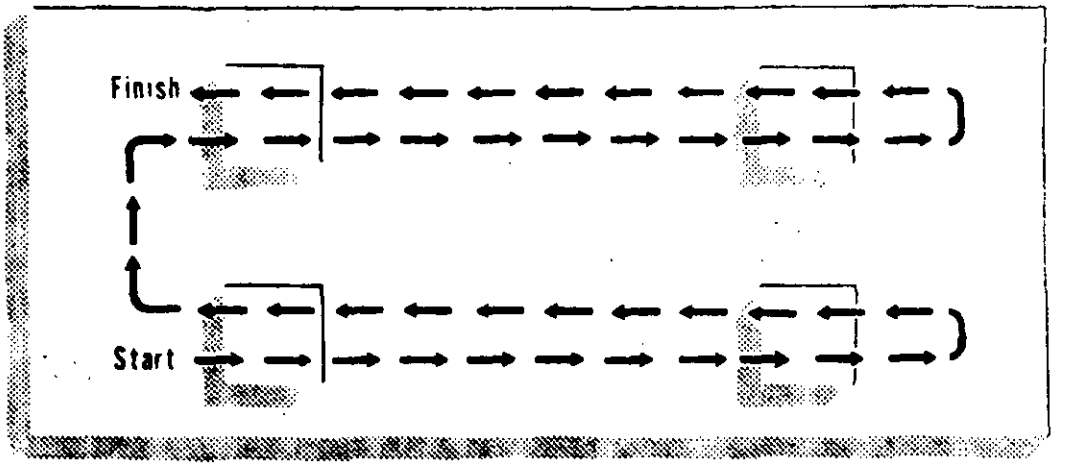
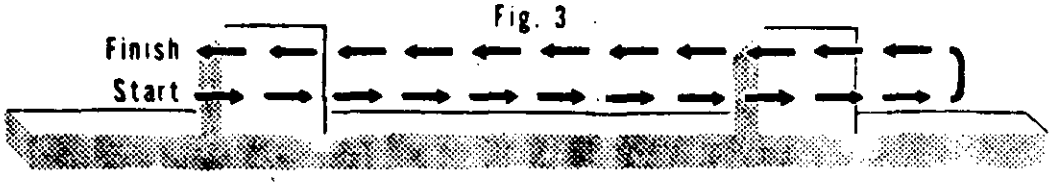


Fig. 3



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* 4.5.4.1 Performance test. A finishing cut shall be made on the surfaces that were previously rough cut under 4.5.4. The spindle speed, table feed rate, depth of cut, and type of cutting tool shall be determined by the supplier. The surfaces of the workpieces after the cut shall show no visual evidence of tool chatter, and the surfaces of each workpiece shall meet the following requirements:

Squareness - The two finished sides shall be square with each other within 0.001 inch per foot, based on total finished dimensions.

Flatness - The two finished milled surfaces shall be flat within 0.001 inch, concave only.

Parallelism - The horizontal and vertical finished dimensions of the workpiece shall not vary more than 0.001 inch over its entire length.

4.5.5 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.14 in accordance with Table II of MIL-STD-461.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in toolrooms, machine shops, and laboratories.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

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a. Title, number and date of this specification.

b. Style, class and size machine required (see 1.2). For Style 2 and Style 5 state whether side rail head shall be on right side or left side.

c. First Article approval, if required (see 3.1).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

e. Automatic lubrication system, if required (see 3.2.3).

f. Painting, if different (see 3.3.5).

g. Coolant drain system, if different (see 3.4.5).

h. Table controls, if different (see 3.4.5.2).

i. Crossrail head movement control, if different (see 3.4.6).

j. Power drawbar for crossrail milling head spindle(s), if different (see 3.4.6.1).

k. Crossrail head quill controls, if different (see 3.4.6.1).

l. Crossrail head spindle speed pendant control for Class 3 machine, if required (see 3.4.6.1).

m. Siderail head power saddle clamp, if required (see 3.4.7).

n. Siderail head pendant control of movement, if required (see 3.4.7).

o. Ram type side head, if required (see 3.4.7).

p. Siderail head spindle quill controls and clamp, if different (see 3.4.7.1).

q. Pendant siderail head quill control, if required (see 3.4.7.1).

r. Siderail head spindle draw-bar, if different (see 3.4.7.1).

s. Siderail head spindle speed pendant control for Class 3 machine, if required (see 3.4.7.1).

t. Specify required table length (see Table I).

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u. Options, if required (see 3.5 and Table I).

v. Electrical system requirements, if different, and voltage for initial wiring, if different (see 3.8).

w. Spindle motor duty rating, and type of enclosure, if different (see 3.8.1).

x. Standard equipment, if different (see 3.11).

y. Optional equipment, if required (see 3.12).

z. Specify additional equipment, if required (see 3.13).

aa. Electromagnetic interference control, if required (see 3.14 and 6.4).

bb. Type fungus proofing, if required (see 3.15).

cc. Lubrication chart, if different (see 3.16).

dd. Nameplate, if different (see 3.17).

ee. First Article inspection , if different (see 4.2).

ff. Quality conformance inspection, if different (see 4.3).

gg. Cutting test material, if different (see 4.5.3).

hh. Electromagnetic interference control test, if different (see 4.5.5 and 6.4).

ii. Required level of preservation-packaging (see 5.1).

jj. Required level of packing (see 5.2).

kk. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

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* 6.4 Army and Air Force procurements. When equipment purchased for the Department of the Army or the Air Force requires electromagnetic interference control testing, such testing should be required to be in accordance with MIL-STD-461, Class IIB equipment, except only Test Methods CE03 and RE02 should be required.

* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.7 Mercury prohibited. When the use of mercury or its compounds in the machine and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
GSA - GS

User Activities:
Army - WC
Navy - MC

Preparing Activity:
DSA-IP

Project Number:
3417-0103

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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